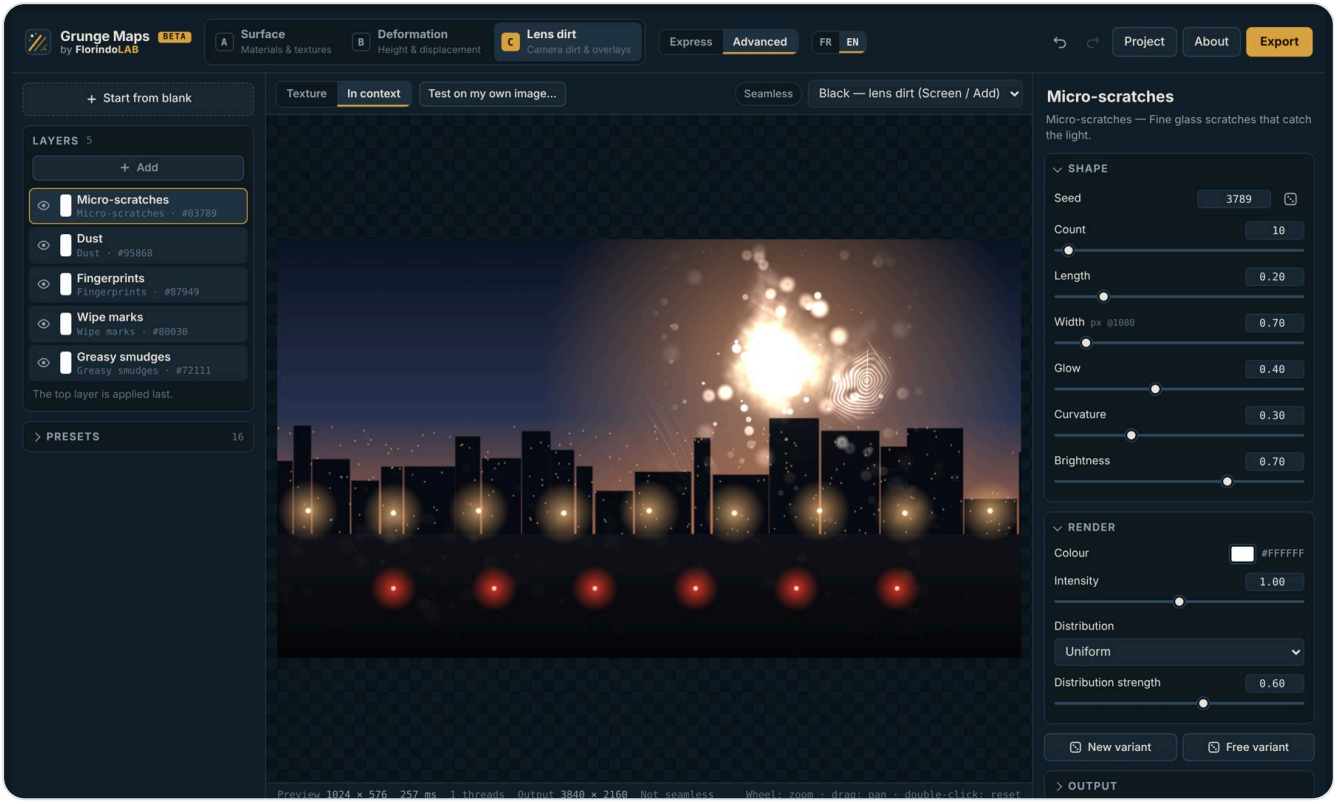
Grunge Maps

by **FlorindoLAB**

Free grunge map generator for 3D software

User guide



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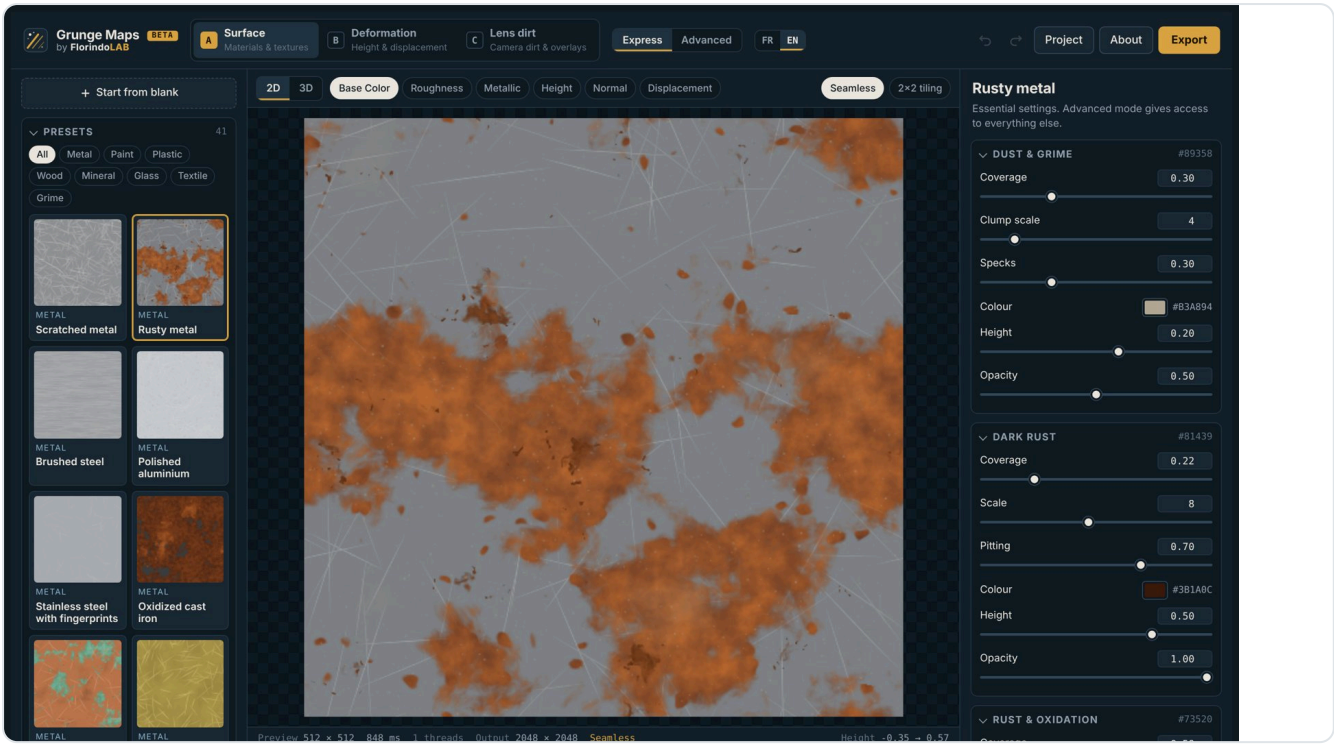
1. Overview

Grunge Maps creates the wear and dirt textures that make a 3D image believable: scratches, dust, rust, cracks, fingerprints, bumps, impacts, and the *lens dirt* that shows up in a camera's highlights. You get maps ready to plug into Blender, Substance, Unreal Engine, Unity, Cinema 4D, or to use as masks in your compositing software.

- **Free, no account, nothing to install**: the tool opens in your browser at **florindolab.com/tools/grunge-maps/**.
- **Everything is computed on your computer.** No image, setting or project is ever sent to a server.
- Recent browsers: Chrome, Edge, Firefox, Safari. A computer is recommended; on a phone the interface still works but is cramped.
- Interface in English and French (**FR / EN** buttons at the top).

> **Beta version.** Presets, settings and the project format may still change. If a map does not behave as expected in your software, or a preset is missing, tell me through **About › Send feedback**.

2. The interface



Overview in Express mode (Surface module, “Rusty metal” preset)

- **Header**: the three modules (**A Surface**, **B Deformation**, **C Lens dirt**), **Express / Advanced** modes, language, Undo / Redo, the **Project** menu, **About** and **Export**.
- **Left panel**: the **Start from blank** button, presets and generators (Express) or the layer stack (Advanced).
- **Centre view**: the 2D or 3D preview, with the bar to choose the displayed map. The status bar at the bottom shows the preview size, computation time, output size and whether the texture is seamless.
- **Right panel**: settings of the selected layer, variants, output and preview.

Express or Advanced

Express shows the essentials: pick a preset or a generator, adjust a few sliders, export. **Advanced** gives access to the whole layer stack and every setting. You can switch at any time without losing anything.

Sliders

Drag a slider to see the result live (fast preview), then release it for the accurate computation. You can also type a value in the box on the right. **Double-click** a slider to reset it to its default value.

3. Presets, variants and blank start

Each module has its own library: **41 presets** in Surface (metal, paint, plastic, wood, mineral, glass, textile, grime), **20** in Deformation and **16** in Lens dirt. Thumbnails show the result; tags filter by category. One click applies the preset.

- **New variant** draws new random seeds and keeps every setting: same material, different draw.
- **Free variant** also slightly changes the secondary settings, to explore.
- **Start from blank** clears the current module. Then add generators one by one: in Express, each click on a generator adds a layer.
- **My presets**: at the bottom of the list, **Save the stack as a preset** keeps the current stack in this browser. It is also saved in the project file.

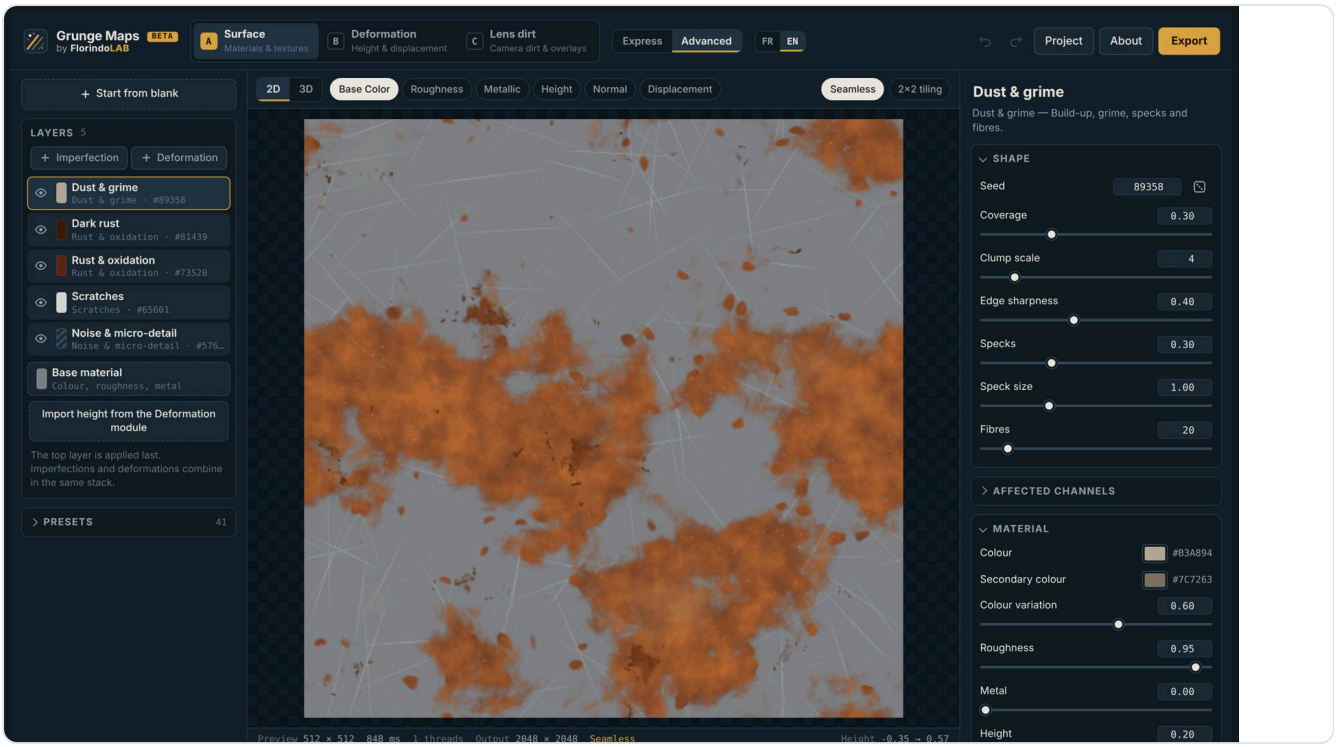
4. Module A · Surface

The Surface module produces a complete PBR material: **Base Color**, **Roughness**, **Metallic**, **Height**, **Normal** and **Displacement**. Layers stack on a **base material** (colour, roughness, metal) that you set by clicking the last row of the stack, in Advanced mode.

Generator	Use it for
Scratches	Random, parallel (brushed metal) or circular (polishing, car-wash marks).
Dust & grime	Grime build-up, specks, fibres.
Rust & oxidation	Patches, pitting, hue variation (also verdigris, lichen...).
Chipped paint	Chips revealing the layer below, with a paint lip.
Stains & drips	Rings, water marks, drips following gravity.
Cracks	Crack network with branches, from fine crackle to dry ground.
Surface irregularities	Broad undulation, stretched grain, roughness variation.
Noise & micro-detail	Material grain (Perlin, ridges, cellular, per-pixel noise).
Wood grain, Weave, Cells	Wood, plain or twill fabric, leather, pebbles, galvanized sheet.
Smudges & prints	Greasy smudges and fingerprints, great for roughness.

Adding relief (Surface + Deformation)

In Advanced mode, two buttons add layers: **+ Imperfection** and **+ Deformation**. A deformation (bumps, hammered, waves...) added in Surface only affects height (Height, Normal, Displacement) and carries a **height** badge. The **Import height from the Deformation module** button copies the whole module B stack at once.



Advanced mode: layer stack and settings of the selected layer

5. Layers in detail

The top layer of the list is applied last. On each row: the eye (show / hide), then on hover a new seed, move up, move down, duplicate and delete. You can also drag a layer to move it, and rename the selected layer by clicking its name in the right panel.

Global scale (above the stack, and at the top of the settings in Express mode) scales every pattern in the stack up or down at once, from 0.1× to 4×, without changing the layer settings. Double-click the slider to go back to 1×. It changes the exported texture; to simply repeat the texture on the object in the 3D preview, use **UV ×** (bar above the preview) or **UV repeat** instead. In seamless mode, very large scales stop at the largest pattern that still tiles.

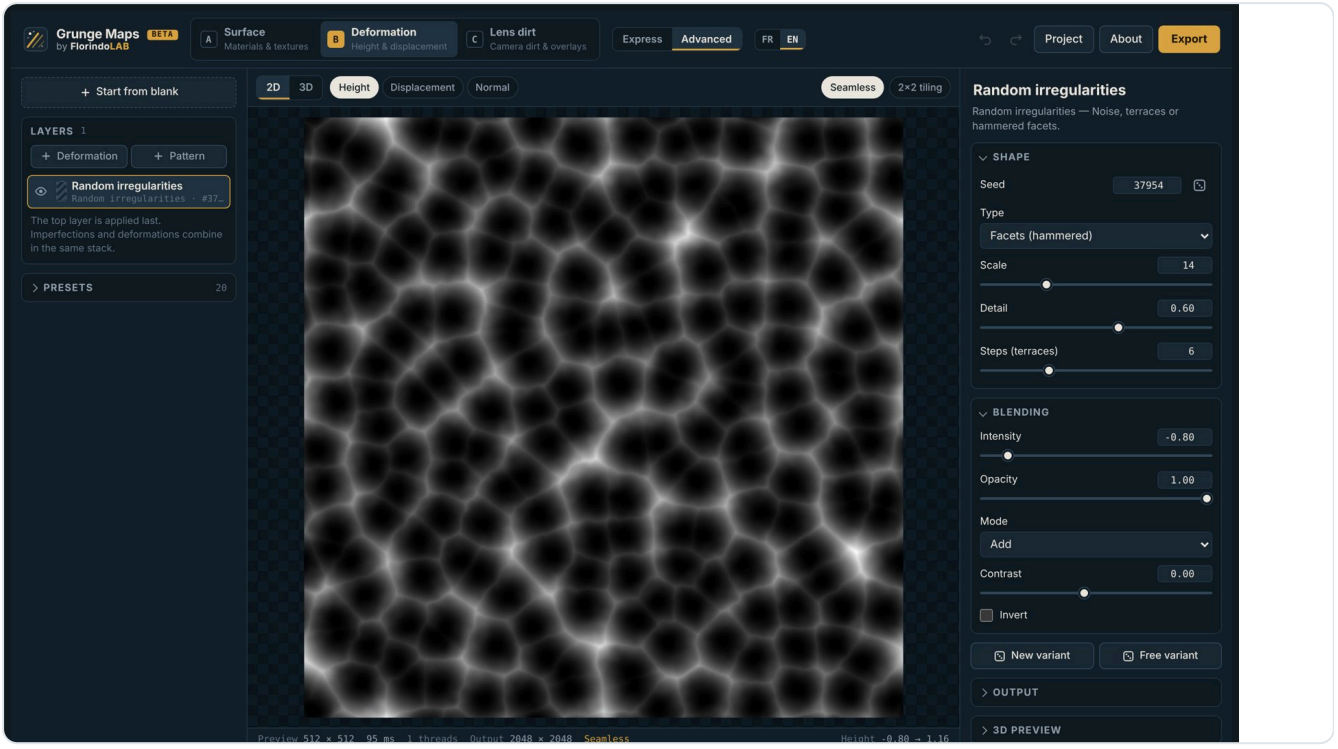
- **Shape**: the seed (the same number always gives the same draw) and the generator's own settings. The "px @1K" unit is a size in pixels for a 1024 px texture: it scales with the export resolution.
- **Affected channels** (Surface): choose whether the layer changes colour, roughness, metal and/or height. A fingerprint layer often affects roughness only.
- **Material**: colour, secondary colour and variation between them, roughness, metal, height (negative = dents).
- **Blending**: opacity, height blend mode (Add, Subtract, Maximum, Minimum, Multiply, Replace), mask contrast and threshold, invert.

6. Module B · Deformation

The Deformation module only produces height: **Height**, **Displacement** and **Normal**. Use it for dented, hammered, wavy, organic or eroded surfaces.

- **Bumps & dents**: domes, craters or cones; dent ratio, size, density.
- **Waves**: sine, triangle, ridges, radial ripples, cross waves.
- **Organic deformations**: fluid shapes, veins or cells.
- **Random irregularities**: noise, terraces, hammered-metal facets.
- **Erosion patterns**: channel erosion and pits.

The **+ Pattern** button also adds the Surface generators (cracks, wood, weave…) converted to height. **Intensity** can be negative to dig instead of raise.



Deformation module: hammered metal height map

7. Module C · Lens dirt

The Lens dirt module generates a full-frame image (1920 × 1080 to 3840 × 2160, or square) showing a dirty lens: blurred dust with bokeh rims, dried drops, greasy smudges, wipe marks, fingerprints, micro-scratches, fibres, haze. It also offers film grain, film dust and a vignette mask.

Background

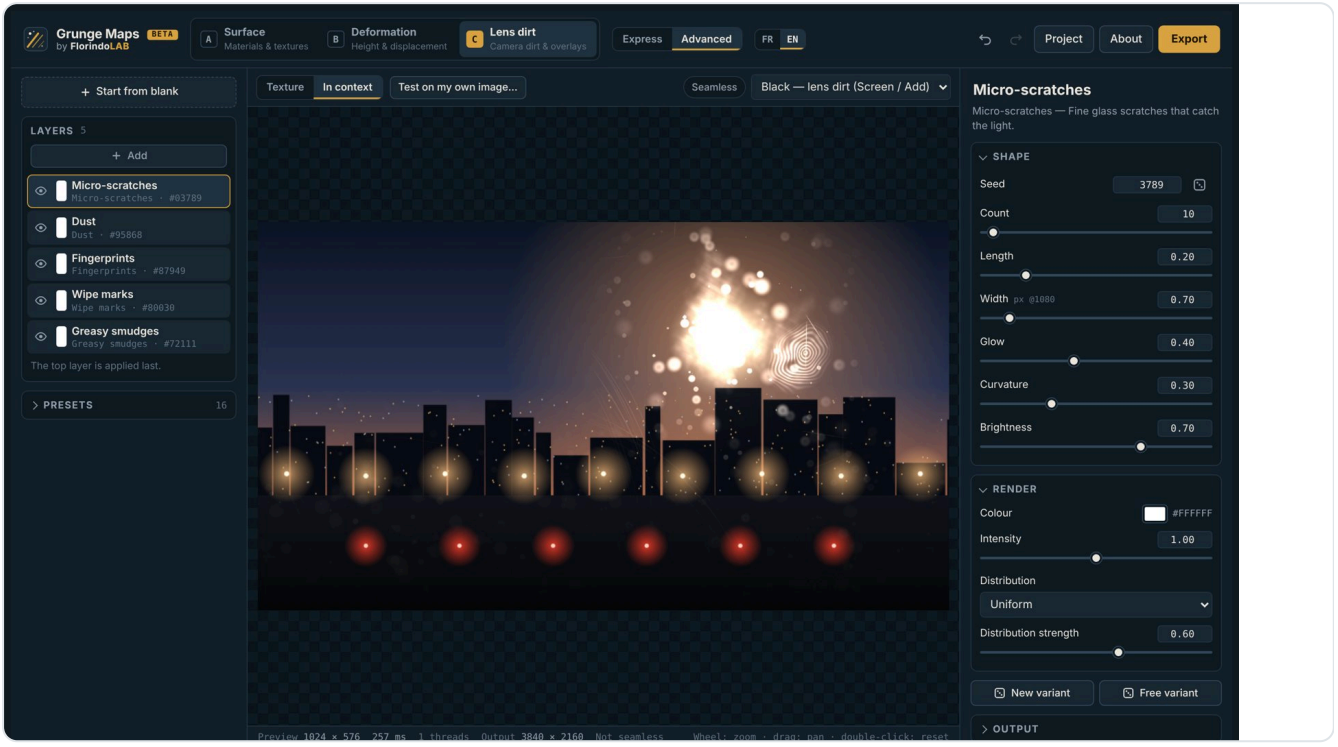
- **Black** (default): classic lens dirt, to use in Screen / Add mode or as a bloom mask.
- **50% gray**: for grain, to place in Overlay mode.
- **White**: for a Multiply mask, such as a dirty print or a vignette.

Distribution

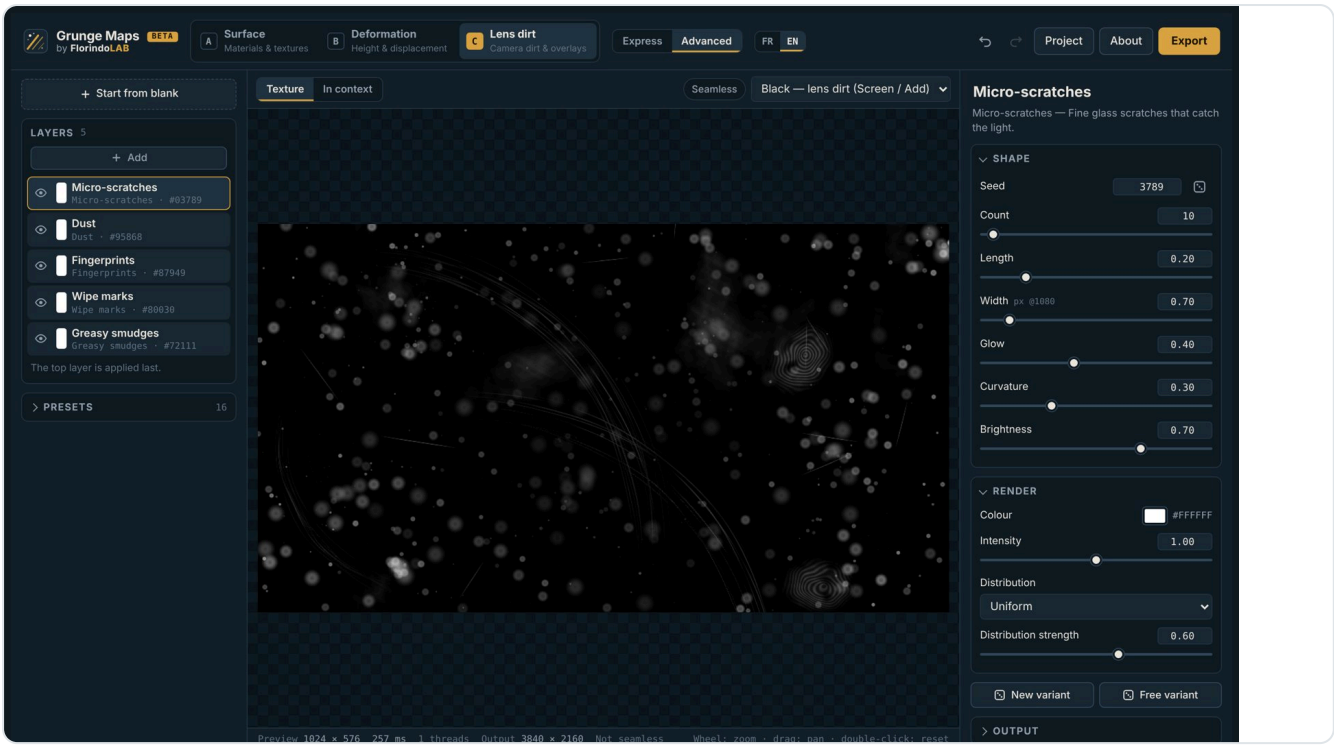
Each layer can gather on the edges of the lens, in the centre, at the bottom (like falling dust) or at the top, with adjustable strength.

Texture or in context

Texture shows the exported image. **In context** simulates use in a 3D application: dirt only shows where the image is very bright. Set the dirt intensity, bloom and threshold in the **In-context preview** panel, and test on your own picture with **Test on my own image…**



“In context” preview: dirt lights up around light sources

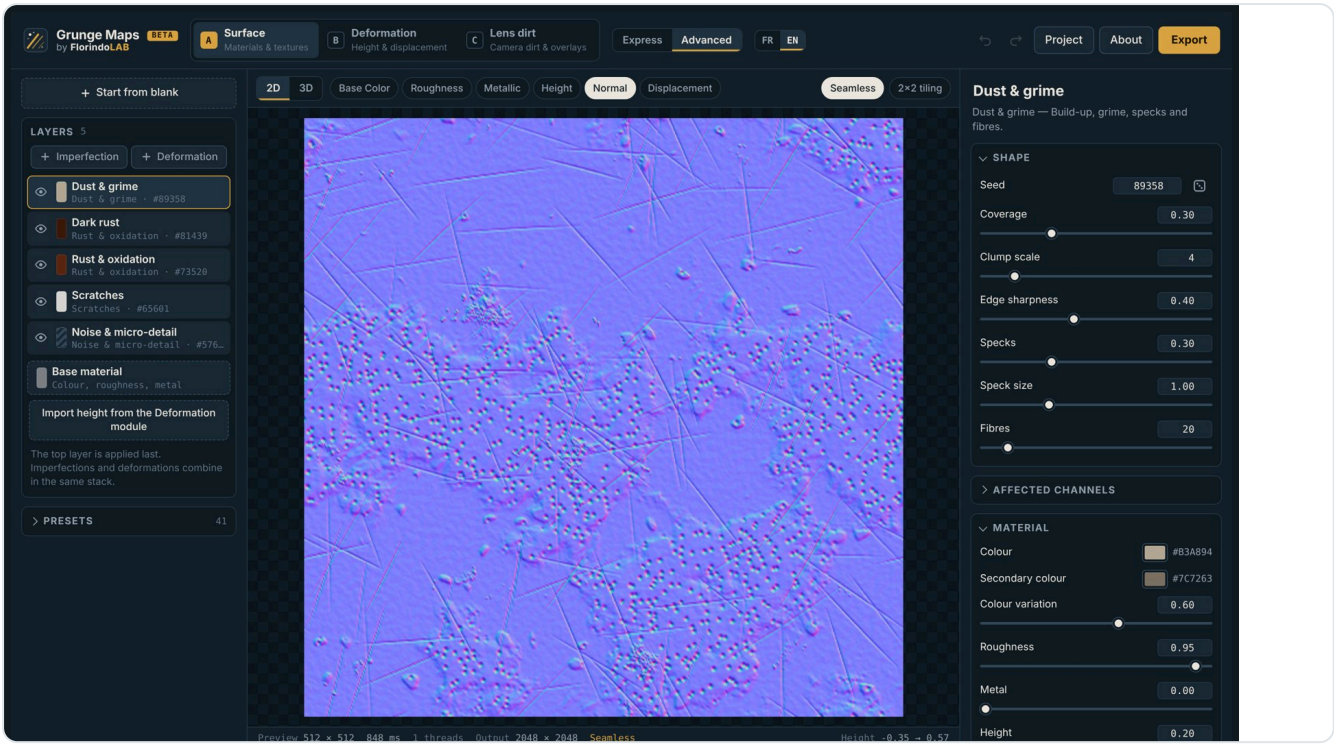


The same texture as exported (black background)

8. 2D and 3D preview

2D

The tags above the view display each map on its own (Base Color, Roughness, Normal…). **2×2 tiling** repeats the texture to check that no seam is visible. Wheel to zoom, drag to pan, double-click to reset.



Displaying the normal map alone

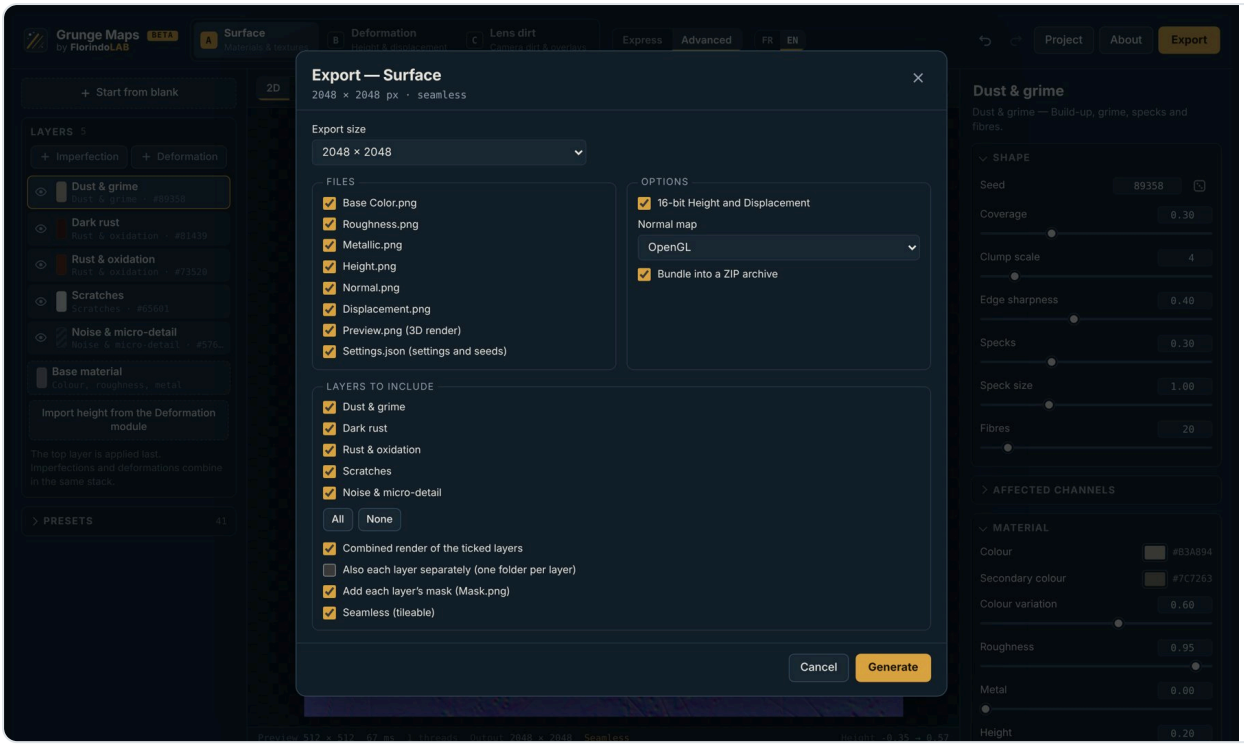
3D

The 3D view applies every map to a sphere, a cube, a plane or **your own model**: choose **My 3D model…** in the object list (.glb recommended, single-file .gltf, .obj, .fbx). Once a model is loaded, **Import another model…** replaces it (same for **Load another HDRI…**). The model should have UVs; otherwise an automatic projection is applied. Displacement only shows on a dense mesh, and the model is not kept between sessions. Drag to rotate, wheel to zoom, double-click to reset the framing. In Advanced mode, the **3D preview** section sets displacement, UV repeat and auto-rotation; UV repeat is also in the bar above the preview (**UV ×**). If patterns look too big on your model, raise it: it is the equivalent of tiling in your 3D software.

Lighting and HDRI

The environment list offers Studio, Outdoor, Sunset and Night. Pick **Custom HDRI…** to light with your own file (.hdr, .exr or equirectangular .jpg / .png panorama). **Show HDRI** displays it behind the object. The **Lighting** section sets intensity, environment rotation and the key light. The HDRI is not kept between sessions: load it again when needed.

9. Export



The export window

- **Size**: 512 to 4096 px square, or a custom size up to 8192 px. Lens dirt: 1080p, 1440p, 4K or square.
- **Files**: tick the maps you need. Height and Displacement can be exported in **16-bit** (recommended for displacement). The normal map follows the **OpenGL** (Blender, Substance) or **DirectX** (Unreal, 3ds Max) convention. Lens dirt: 16-bit grayscale PNG, or 8-bit colour.
- **Layers to include**: choose which layers to export (for instance only scratches and bumps). **Combined render** merges the ticked layers; **Also each layer separately** creates one folder per layer, with a black-and-white **Mask.png** in Surface.
- **Seamless**: tileable texture (the ZIP name then ends with _seamless).
- **Settings.json** keeps settings and seeds to reproduce the result; **Preview.png** adds a 3D render or the in-context preview.

Generate starts the computation, with a progress bar and a cancel button. Check the thumbnails, then download the ZIP or each file separately. In Express, **Quick export** does it all in one click. At 4096 px, allow a few seconds per layer; if the browser runs out of memory, a message suggests a smaller size.

10. Using the maps in your software

Blender (Cycles or Eevee)

- Base Color: Image Texture node in **sRGB**, into Base Color of the Principled BSDF.
- Roughness and Metallic: Image Texture in **Non-Color**, into the matching inputs.
- Normal: Image Texture in **Non-Color** → **Normal Map** node → Normal input. Export with the OpenGL convention.
- Displacement: Image Texture in Non-Color → **Displacement** node (Midlevel 0.5) → material Displacement output. In Cycles, set the material to **Displacement and Bump** and subdivide the mesh.
- Lens dirt: in the compositor, isolate the highlights (**Glare** node, Fog Glow), multiply the result by the lens dirt image (Mix node in **Multiply**), then add it back to the image (Mix in **Add**).

Unreal Engine

- Export the normal map with the **DirectX** convention. On import, Roughness, Metallic and Height must have **sRGB unticked**.
- Lens dirt: in a **Post Process Volume**, under Lens › Bloom, set the texture as **Dirt Mask** and adjust **Dirt Mask Intensity**.

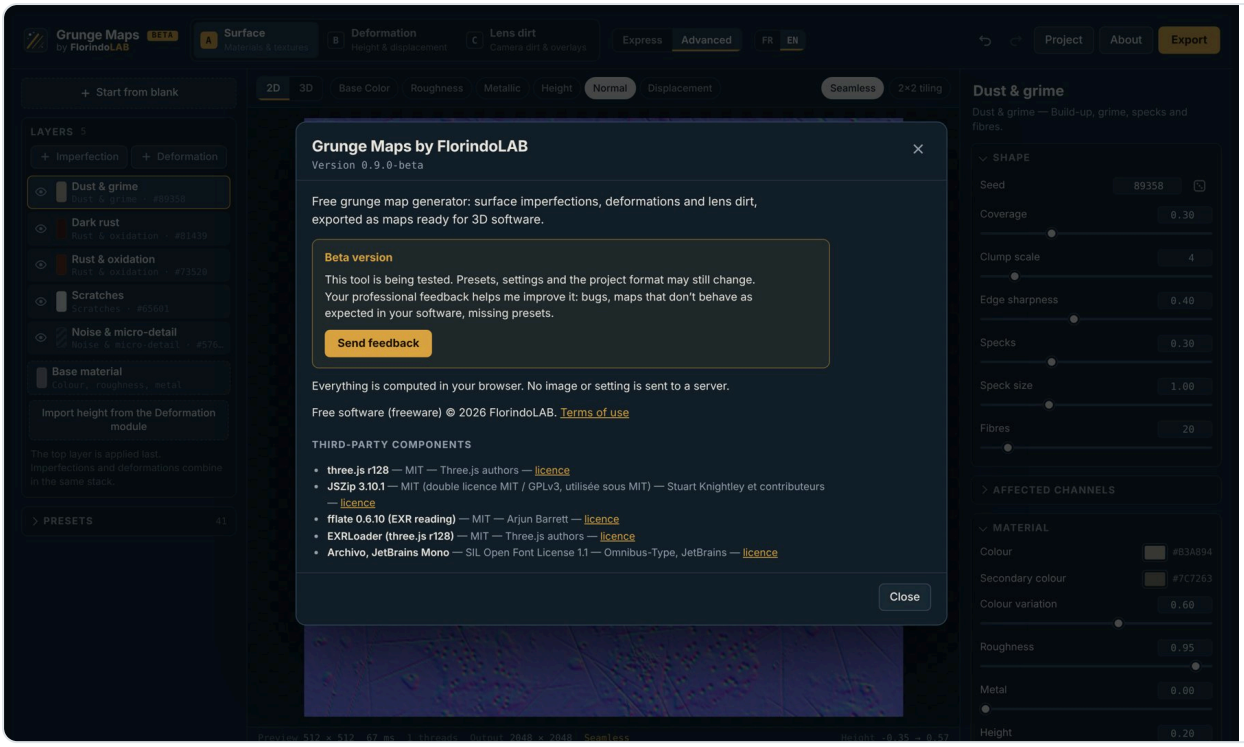
Substance, compositing and motion design

- Grunge maps are often used as **masks**: import a Mask.png or a grayscale map and use it as a layer or fill mask.
- Lens dirt in After Effects, Resolve or Nuke: place the image over the shot in **Screen / Add** mode, ideally masked by the shot's highlights. Grain on a gray background goes in **Overlay**.

11. Projects, undo and shortcuts

- Your work is **saved automatically** in this browser. It is lost if you clear the site data, in private browsing, or on another computer.
- **Project › Save project (.json)** creates a file to keep or share; **Open project…** loads it back. The file contains the three modules, settings and your custom presets, but not imported images or the HDRI.
- **Project › New project** starts again from the default presets; **Clear current module** is the same as a blank start.
- Shortcuts: **Ctrl+Z** undo, **Ctrl+Shift+Z** or **Ctrl+Y** redo, **Esc** closes dialogs.

12. Feedback, licence and privacy



The About window

- **Feedback**: the **Send feedback** button opens the contact form at **florindolab.com/contact/**. Mention your browser and 3D software and, if possible, attach the Settings.json or project file.
- **Licence**: free software (freeware) © 2026 FlorindoLAB. The maps you create are yours and you can use them in your projects, including commercial ones. Full terms: **florindolab.com/licence/**.
- **Privacy**: all computation happens in your browser; nothing is sent to a server.
- **Third-party components**: three.js (MIT), JSZip (MIT), fflate (MIT), Archivo and JetBrains Mono fonts (SIL OFL 1.1). The full list is in the About window.